Missouri Flood Impact Explorer: App Guide

CSE 472 – Project 2

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Purpose

The Missouri Flood Impact Explorer is a Google Earth Engine app. It shows the data behind the flood impact model and the model's predictions, for each of Missouri's 115 counties and each month from 2017 to 2023. The model predicts monthly flood property loss so that counties can be ranked for emergency resources. This guide summarizes the model inputs, then explains what the map shows for each choice in the app.

1 Summary of the Reduced Inputs

The model uses **12 inputs plus calendar month**. Each input stands for one way that flooding causes loss. State is also used in the full Midwest model, but it drops out here because every county is in Missouri.

Block	Variable	Meaning	Source
Hazard, same month	`p_month`	Total rain in the month	PRISM
	`rx5_rl_ratio`	Wettest 5 days ÷ the county's 10-year 5-day level	PRISM
	`snowmelt_mm`	Snowmelt in the month	ERA5-Land
	`frost_days`	Days with minimum temperature below 0 °C	PRISM
Hazard, day 1	`sm_root_pct`	Soil wetness on day 1, as a percentile against 1950–1999	ERA5-Land
	`swe_mm`	Snowpack on day 1 of the month	ERA5-Land
Rain climate	`c_rx5_rl10_mm`	10-year 5-day rain level, from 2000–2009	PRISM
Exposure and susceptibility	`log_built_surface`	Built-up area, natural log, previous year	GHSL
	`impervious_mean`	Paved share of the county, previous year	Annual NLCD
	`hand_lt5_share`	Share of land less than 5 m above nearest drainage	MERIT Hydro
	`log_land_km2`	County land area, natural log	Census TIGER
Adaptation	`wetland_share`	Wetland share of the county, previous year	Annual NLCD

The model has two parts: one predicts the chance of any damage, and the other predicts how large the loss would be if damage occurs. **Expected loss** is the two multiplied together, and counties are ranked by that number.

2 How to Use the App

Control	What it does
First Layer menu	Picks a group: predictions, county inputs, or a source dataset
Second Layer menu	Picks one layer within that group
Year and Month menus	Pick the month, from January 2017 to December 2023
Previous and Next	Move one month at a time
Outline checkbox	Draws a cyan outline around the 6 highest-risk counties that month (the top 5%)
Colour bar	Shows the colour range and units of the current layer
Click on a county	Outlines it in black and opens its details in the side panel

The app opens on **Model predictions, Expected loss, April 2017**, Missouri's largest damage month in the test years.

Two kinds of map

- **County maps** give each county one colour. These are the values the model used, or the model's predictions.
- **Source maps** show the original gridded data inside Missouri, with county borders on top. Colour can vary within a county.

Values beyond the ends of the colour bar are drawn in the end colour.

3 What the Map Shows for Each Choice

The last column of each table shows what makes the map change. "Fixed" means the map stays the same when the date changes.

3.1 Model predictions

Layer	What the map shows	Colour range	Changes by
Expected loss	Predicted dollars of loss in that county and month. Darker red is a higher predicted loss. This is the main heatmap.	\$100 to \$10M, log scale	Month
Chance of any damage	Probability that the county records any flood damage	0% to 60%	Month
Rank within the state	The county's position among all 115. Darkest is rank 1, the highest expected loss.	1 to 115	Month
Recorded loss (SHELDUS)	The loss that was actually recorded, for comparison. Blank counties have no record that month.	\$1k to \$100M, log scale	Month

3.2 Model inputs (county values)

These are the exact values given to the model for that county and month.

Layer	What the map shows	Colour range	Changes by
Monthly rain	`p_month`: county-average rain total	0 to 300 mm	Month
Wettest 5 days / 10-year level	`rx5_rl_ratio`: how extreme the month's heaviest 5 days were. A value of 1 equals a 10-year event.	0 to 1.3	Month
Snowmelt	`snowmelt_mm`: melt during the month	0 to 35 mm	Month
Frost days	`frost_days`: days with minimum temperature below 0 °C	0 to 30 days	Month
Soil wetness percentile	`sm_root_pct`: how wet the soil was on day 1, compared with the same month in 1950–1999	0% to 100%	Month
Snowpack	`swe_mm`: water held in snow on day 1	0 to 15 mm	Month
10-year 5-day rain	`c_rx5_rl10_mm`: the county's long-run heavy-rain level	125 to 210 mm	Fixed
Built-up surface (log)	`log_built_surface`: amount of built-up area, previous year	14.5 to 18.6	Year
Paved share	`impervious_mean`: paved percentage of the county, previous year	0% to 25%	Year
Land under 5 m above drainage	`hand_lt5_share`: share of the county that sits low near water	10% to 100%	Fixed
County land area (log)	`log_land_km2`: size of the county	6.5 to 7.9	Fixed
Wetland share	`wetland_share`: wetland share of the county, previous year	0% to 8.5%	Year

3.3 Source datasets

Group	Layer	What the map shows	Colour range	Changes by
PRISM	Monthly rain	Sum of daily rain	0 to 300 mm	Month
	Wettest single day	Largest one-day rain in the month	0 to 100 mm	Month
	Frost days	Days with minimum temperature below 0 °C	0 to 31 days	Month
ERA5-Land	Root-zone soil water	Soil water from 0 to 100 cm on day 1	0.15 to 0.45 m^3/m^3	Month
	Snowpack	Snow water on day 1; blank where there is none	0 to 30 mm	Month
	Snowmelt	Monthly melt; blank where there is none	0 to 40 mm	Month
GHSL	Built-up surface	Built area in each 100 m cell; blank where there is none	0 to 3,000 m^2	Year*
NLCD	Land cover	16 land-cover classes, listed in the panel	Classes	Year
	Impervious surface	Paved percentage; blank where there is none	0% to 80%	Year
	Wetlands	Wetland areas in blue	Yes only	Year
Terrain	Elevation	Height of the land	70 to 500 m	Fixed
	Slope	Steepness of the land	0 to 15°	Fixed
	Height above drainage	Height above the nearest stream. Darker is closer to stream level.	0 to 30 m	Fixed
	Land under 5 m above drainage	Low land near streams in blue	Yes only	Fixed

NLCD and GHSL use the previous year's data, so that the inputs are known before the month being predicted. *GHSL only has 5-year images: the map shows the 2015 image for 2017–2018 and the 2020 image for 2019–2023. The county input is interpolated between images, so it can differ.

How a county input relates to its source map

Source map	County input in the model
PRISM Monthly rain	County average of the same data (`p_month`)
PRISM Frost days	Days calculated from the county-average daily temperature (`frost_days`)
ERA5-Land Root-zone soil water	Converted to a percentile against 1950–1999 (`sm_root_pct`)
ERA5-Land Snowpack, Snowmelt	County average (`swe_mm`, `snowmelt_mm`)
GHSL Built-up surface	County total, then natural log (`log_built_surface`)
NLCD Impervious surface	County average (`impervious_mean`)
NLCD Wetlands	County share (`wetland_share`)
Terrain: Land under 5 m above drainage	County share (`hand_lt5_share`)
PRISM Wettest single day; NLCD Land cover; Terrain Elevation, Slope, Height above drainage	Shown for context. These layers are not model inputs themselves.

4 What Clicking a County Shows

- **Prediction:** chance of any damage, expected loss, loss if damage occurs, rank in the state, and recorded loss.
- **Model inputs:** the 12 values for that month. Built-up surface and land area are converted from logs to km^2 so they are easier to read.
- **County mean:** for most source maps, an extra row gives that map's average over the county. Land cover and the two yes-or-no maps (Wetlands and Land under 5 m) do not have one.
- **Chart:** every month from 2017 to 2023. The red line is the predicted expected loss and purple dots are recorded losses. The axis is in $\log_{10}$ dollars, so 3 is \$1k and 6 is \$1M.

The details update when the month or layer changes.

5 Worked Example: April 2017

With **Model predictions, Expected loss, April 2017**, the darkest counties are in southern Missouri. The 6 outlined counties are below, and each had recorded damage. Together they hold 63% of the state's \$150M recorded loss that month. In all, 35 of the 115 counties recorded damage.

County	Rank	Chance of damage	Expected loss	Recorded loss
Barry	1	82%	\$10.8M	\$1.3M
Taney	2	83%	\$9.7M	\$12.8M
Greene	3	76%	\$9.6M	\$0.6M
Phelps	4	90%	\$8.4M	\$1.3M
Howell	5	87%	\$8.2M	\$76.9M
Newton	6	78%	\$8.2M	\$1.3M

The model picked the right area, but the dollar amounts differ widely. Howell recorded about 9 times its expected loss, and Greene about one-fifteenth of it.

To see why, keep April 2017 and try these combinations:

Choice	What to look for
Model predictions, Recorded loss	Compare recorded damage with the predicted pattern
Model inputs, Monthly rain	The heaviest county rain; Howell had 430 mm
Model inputs, Wettest 5 days / 10-year level	Values above 1 mean worse than a 10-year event; Howell reached 1.58
PRISM, Monthly rain	Where within counties the rain fell
Terrain, Land under 5 m above drainage	Where low land near streams sits
GHSL, Built-up surface	Where buildings are concentrated

6 Limits to Keep in Mind

- **Predictions are computed outside Earth Engine.** LightGBM cannot run in Earth Engine, so the predictions are uploaded as a table and do not change with the map.
- **They are not true forecasts.** Predictions use the rain that actually fell that month.
- **Compare counties more than dollar amounts.** Dollar estimates are typically off by several times. Missouri's top 5% is only 6 counties, so one large month can swing results.
- **Only 2017–2023 is shown.** The model was fitted on 2000–2016, so these are the years it did not train on.
- **Blank does not always mean zero loss.** A county with no SHELDUS record may still have had unreported damage.
- **NFIP flood insurance and buyouts are not included.** That data is still being collected.

7 How the Predictions Are Calculated

The app does not calculate predictions itself. It looks up a row in `mo_flood_app_table.csv`, which is built ahead of time by `build_app_table.py` using the LightGBM hurdle model in `reduced_benchmark.py`. This section follows one county through that calculation: **Miller County, April 2017**.

The two models

Both models are trained only on Missouri counties for 2000–2016, so April 2017 is out of sample.

	Part 1: classifier	Part 2: regressor
Predicts	Chance of any damage that month, p	$\log_{10}$ of the loss, assuming damage happened, μ
Trained on	All county-months	Only county-months with damage
Miller County	$p = 0.375$	Not saved in the table

The number of trees is set by early stopping (fit on 2000–2012, validate on 2013–2016), then multiplied by 1.15 for the final fit on all of 2000–2016.

Inputs for Miller County, April 2017

Input	Value
Monthly rain	250.8 mm
Wettest 5 days / 10-year level	1.24 (above a 10-year event)
Snowmelt, snowpack, frost days	About 0
Soil wetness percentile	72%
10-year 5-day rain	142.6 mm
Log built-up surface	16.02
Paved share	1.2%
Land under 5 m above drainage	18%
Log land area	7.34
Wetland share	0.35%
Calendar month	April

Calendar month is a 13th input that the side panel's input list does not show.

How each number in the side panel is produced

The expected loss combines both models:

$$\text{Expected loss} = p \times 10^{\mu} \times \text{smear}$$

The smear factor is a single correction: the average of $10^{(\text{actual}-\text{predicted})}$ over the training months with damage. Without it, converting from $\log_{10}$ back to dollars would give something closer to a typical loss and understate the average.

Side panel field	Calculation	Miller County
Chance of any damage	p from Part 1	0.3753, shown as 38%
Expected loss	$p \times 10^{\mu} \times$ smear	\$1,377,640, shown as \$1.4M
Loss if damage occurs	Expected loss $\div\ p$, which equals $10^{\mu} \times$ smear	\$3,670,660, shown as \$3.7M
Rank in the state	Position by expected loss among the 115 counties that month	40
Recorded loss (SHELDUS)	Actual inflation-adjusted loss; not a prediction or an input	\$2,563,200, shown as \$2.6M

Reading these numbers

- **Compare \$3.7M to the recorded loss, not \$1.4M.** Damage did occur, so the loss if damage occurs is the like-for-like estimate. Here it is about 1.4 times too high. The \$1.4M is an average that includes the 62% chance of no damage.
- **Loss if damage occurs is closer to a mean than a typical value.** The smear factor pulls it toward rare, very large losses, so a typical damaging month would be lower.